

MedTrack DV - Hospital Operations & Patient Analytics

Module 4: Dashboard Storyboard & Architecture Specification

Project Name: MedTrack DV (Hospital Operations & Patient Analytics Dashboard)

Deliverable Document: dashboard_storyboard.pdf / Specification Guide

Milestone: Milestone 2 - Module 4 (Dashboard Planning & Prototyping)

1. Executive Summary & Architecture Overview

This document serves as the official storyboard and structural framework specification for the MedTrack DV Tableau Analytics Suite. Designed for hospital administrators, department heads, and operational managers, the dashboard suite transforms complex healthcare transactional data into interactive, actionable visual analytics.

The unified Tableau workbook (medtrack_prototype.twbx) connects directly to the processed dataset hospital_final_dataset.xlsx containing 55,500 patient records across 26 metrics. The architecture is structured into four interconnected analytical dashboards:

1. **Hospital Overview:** Strategic C-suite summary of hospital volume, occupancy, readmissions, and monthly operational trends.
2. **Patient Flow Analytics:** Tactical tracking of admission paths, discharge throughput, capacity bottlenecks, and length of stay (LOS).
3. **Department Analytics:** Operational evaluation of department volumes, waiting times, efficiency scores, and clinical performance.

4. **Resource Utilization:** Capacity planning dashboard tracking bed utilization, physician workload, nurse allocation, and critical equipment.

2. Core Healthcare KPIs & Mathematical Formulas

The following core KPI calculated fields have been engineered and validated within Tableau for implementation across the top banner of all dashboards:

KPI Field Name	Tableau Calculated Field Formula	Validated Benchmark Value	Business Significance
Total Admissions	COUNT([patient_id])	55,500	Measures overall hospital demand and total volume of served patients.
Occupancy Rate (%)	SUM(FLOAT([occupied_beds])) / SUM(FLOAT([total_beds]))	76.71%	Monitors capacity strain and overall bed availability percentage.
Average Length of Stay (LOS)	AVG([length_of_stay_days])	4.87 Days	Evaluates inpatient care efficiency and bed turnover rate.
Readmission Rate (%)	SUM(FLOAT([is_readmission])) / COUNT([patient_id])	8.09%	Key clinical quality metric tracking 30-day post-discharge return rates.
Bed Utilization	(SUM(FLOAT([occupied_beds])) / SUM(FLOAT([total_beds])))	76.71%	Tracks physical

KPI Field Name	Tableau Calculated Field Formula	Validated Benchmark Value	Business Significance
Rate (%)	$\text{pied_beds}})) / \text{SUM(FLOAT([total_beds]))} * 100$		infrastructure deployment efficiency across wards.
Dept Wait Time Score	$\text{AVG([avg_wait_time_hours])}$	2.75 Hours	Operational efficiency KPI measuring pre-admission patient waiting times.

3. Detailed Dashboard Layout & Wireframe Specifications

Dashboard 1: Hospital Overview Wireframe

Target Audience: Chief Executive Officer (CEO), Chief Medical Officer (CMO), Board Members.

Layout Grid & Structure:

- **Top KPI Banner:** 6 Executive Scorecard Tiles displaying Total Admissions (55,500), Occupancy Rate (76.71%), Avg LOS (4.87 Days), Readmission Rate (8.09%), Bed Utilization (76.71%), and Total Discharges.
- **Middle Row (Trend Analytics):**
 - Line Chart 1: Monthly Admissions Trend (12 Months).
 - Line Chart 2: Monthly Occupancy Rate Trend (%) vs Threshold Target (80%).
 - Line Chart 3: Monthly Readmission Rate Trend (%).
- **Bottom Row (Breakdowns & Regional Mapping):**

- Donut Chart: Admissions by Patient Type (Inpatient, Outpatient, Emergency, Day Care).
- Bar Chart: Admissions by Department (General Medicine, Surgery, Pediatrics, Cardiology, Orthopedics, ICU, Emergency).
- Bar Chart: Avg LOS by Department (Days).
- Horizontal Bar Chart: Readmission Rate by Department (%).
- Geographic / Regional Map: Hospital Admissions distribution across regions.
- Combo Chart: Monthly Admissions vs Discharges throughput.

Dashboard 2: Patient Flow Analytics Wireframe

Target Audience: Patient Flow Managers, Emergency Ward Supervisors, Chief Nursing Officers.

Layout Grid & Structure:

- **Top Summary Bar:** Peak Daily Inflow, Avg Discharge Processing Time, Overnight Transfer Rate, Capacity Strain Index.
- **Main Visualizations:**
 - Sankey / Flow Diagram (or Grouped Bar): Admission Source (ER, Referral, Direct, Transfer) -> Department -> Discharge Status.
 - Heatmap: Patient Inflow by Day of Week vs Hour of Day (Identifying peak ER rush hours).
 - Box Plot / Histogram: Length of Stay (LOS) Distribution across Medical Conditions.
 - Dual-Axis Area Chart: Daily Cumulative Admissions vs Daily Cumulative Discharges.

Dashboard 3: Department Analytics Wireframe

Target Audience: Department Chairs, Clinical Directors, Operations Research Team.

Layout Grid & Structure:

- **Top Summary Bar:** Busiest Department, Highest Wait Time Dept, Department Efficiency Index.
- **Main Visualizations:**
 - Scatter Plot: Patient Volume (X-axis) vs Avg Wait Time in Hours (Y-axis) sized by Readmission Rate.
 - Stacked Bar Chart: Treatment Capacity vs Occupied Beds by Department.
 - Side-by-Side Bar Chart: Scheduled vs Emergency Admissions per Department.
 - Metric Table: Departmental Scorecard (Admissions, Avg Billing, Wait Time, Mortality Rate, Readmission %).

Dashboard 4: Resource Utilization Wireframe

Target Audience: Hospital Operations Director, Resource Planning Manager, Supply Chain Head.

Layout Grid & Structure:

- **Top Summary Bar:** Total Hospital Beds (3,467,385), Total Occupied Beds (2,659,683), Available Beds (807,702), Nurse-to-Patient Ratio.
- **Main Visualizations:**
 - Bullet Graph: Bed Occupancy vs Maximum Capacity Thresholds per Ward.
 - Heatmap Matrix: Physician & Nursing Staff Workload Allocation across Shift Hours.
 - Bar Chart: Medical Equipment & ICU Ventilator Utilization Rate (%).
 - Gantt / Timeline View: Scheduled Maintenance & Bed Turnover Downtime.

4. Interactive Architecture & Filter Integration

To deliver an intuitive user experience, global inter-dashboard navigation and cross-filtering rules are defined as follows:

Interactive Component	Implementation Mechanism	Scope & Functionality
Global Date Range Filter	Tableau Date Parameter / Range Slider	Applies across all 4 dashboards to slice data by Admission Date (2024–2025).
Hospital / Region Filter	Single/Multi-Select Dropdown Filter	Filters entire workbook metrics by specific hospital facilities or geographic zones.
Department Filter	Categorical Dropdown Filter	Isolates metrics for

Interactive Component	Implementation Mechanism	Scope & Functionality
		Cardiology, ICU, Orthopedics, General Medicine, etc.
Cross-Filter Actions	Tableau Dashboard Action (Filter)	Clicking a specific Department bar on Dashboard 1 dynamically updates all surrounding charts.
Navigation Menu	Custom Tableau Button / Storyboard Tab	Allows instant 1-click seamless jumps between Overview, Patient Flow, Depts, and Resources.

5. Deliverables & Next Steps Checklist

- **Workbook Prototype File:** medtrack_prototype.twbx generated with 6 verified calculated fields and 4 structured dashboard canvas frames.
- **Storyboard Document:** dashboard_storyboard.pdf detailing layout wireframes, filter rules, and metric definitions.